Danmarks Tekniske Universitet

Advanced Engineering Mathematics 2
2025 Exam Notes

Olgierd Wladyslaw Janik

1 Differential Equations And Systems

1.1 Linear n^{th} Order Differential Equations

An n^{th} order Hom. Lin. Diff. Eq has n linearly independent solutions $y_1, \ldots, y_n$ that can be written as $y = c_1 y_1 + c_2 y_2 + \ldots + c_n y_n$

A Diff. Eq. can be represented as a polynomial of λ's with power equal to the order. Ex:

$$y''' + 4y'' + 2y' + y = \lambda^3 + 4\lambda^2 + 2\lambda + 1$$

Solving this polynomial gives solutions $e^{\lambda_n t}$ to the Diff. Eq, written as $y = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} + \ldots + c_n e^{\lambda_n t}$

If there is a repeating root, a t is added for every repetition. Ex:

$$(\lambda - 2)^2 \rightarrow c_1 e^{2t} + c_2 t e^{2t}$$

If a solution to the polynomial is complex, you can find two real solutions as $e^{at}cos(wt)$ and $e^{at}sin(wt)$ where $\lambda = a \pm wi$)

$$(\lambda^2 + 4) \rightarrow \lambda = 0 \pm 2i \rightarrow c_1 cos(2t) + c_2 sin(2t)$$

For an inhomogeneous Diff. Eq, either guess a sensible answer and test it, or use the transfer function H(s), which is the polynomial for the Right Hand Side (RHS) over the LHS

$$y''' + 2y'' + y' = 3u'' + 5u', H(s) = \frac{3s^2 + 5s}{s^3 + 2s^2 + s}$$

If $u(t) = e^{st}$, where s is now given, the solution is $H(s)e^{st}$

If $u(t) = cos(wt)$, the solution is $Re(H(iw)e^{iwt})$

If $u(t) = sin(wt)$, the solution is $Im(H(iw)e^{iwt})$ Ex:

$$H(s) = \frac{3s^2 + 5s}{s^3 + 2s^2 + s}, u(t) = cos(t), w = 1$$

$$Re(\frac{3i^2 + 5i}{i^3 + 2i^2 + i}e^{iwt}) = Re((\frac{3}{2} - \frac{5}{2}i)e^{iwt}) = \frac{3}{2}cos(t) - (-\frac{5}{2}sin(t))$$

The real part gets "assigned" to cos(wt), and the imaginary part to sin(wt), and in the form cos(t)-sin(t)

For u(t) = sin(t), the real part gets "assigned" to sin(t) and the imaginary part to cos(t), in the form sin(t) + cos(t)

$$H(s) = \frac{3s^2 + 5s}{s^3 + 2s^2 + s}, u(t) = sin(t), w = 1 \rightarrow \frac{3}{2}sin(t) + (-\frac{5}{2}cos(t))$$

1.2 Systems of Differential Equations & Stability

For a Hom. System, the λ are the eigenvalues of the matrix A where $\underline{x}'(t) = Ax(t)$. The solutions have the form $c_n e^{\lambda_n t} v_n$ where v_n is the corresponding eigenvector.

For a complex pair we get two solutions $Re(e^{\lambda t}v)$ and $Im(e^{\lambda t}v)$. This is effectively the same procedure as with H(s) of "assigning" the real and imaginary parts of the eigenvector to cos or sin depending on Re() or Im() Ex:

$$\lambda = \pm i, w = 1, v_1 = \begin{bmatrix} \frac{-1-i}{2} \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} \frac{-1+i}{2} \\ 1 \end{bmatrix}$$

$$x_1 = cos(t)\begin{bmatrix} \frac{-1}{2} \\ 1 \end{bmatrix} - sin(t)\begin{bmatrix} \frac{-1}{2} \\ 0 \end{bmatrix}, x_2 = sin(t)\begin{bmatrix} \frac{-1}{2} \\ 1 \end{bmatrix} + cos(t)\begin{bmatrix} \frac{-1}{2} \\ 0 \end{bmatrix}$$

$$x(t) = c_1(cos(t)\begin{bmatrix} \frac{-1}{2} \\ 1 \end{bmatrix} - sin(t)\begin{bmatrix} \frac{-1}{2} \\ 0 \end{bmatrix}) + c_2(sin(t)\begin{bmatrix} \frac{-1}{2} \\ 1 \end{bmatrix} + cos(t)\begin{bmatrix} \frac{-1}{2} \\ 0 \end{bmatrix})$$

In the case of a repeated root, find the eigenvector v_1 for the repeated eigenvalue λ, then solve for v_2 the equation:$(A-\lambda I)v_2 = v_1$, where $(A-\lambda I)$ is the matrix A modified by the eigenvalue. Then the solution for this repeated root is $c_1 e^{\lambda t} v_1 + c_2 e^{\lambda t} v_2 + c_2 t e^{\lambda t} v_1$

To solve an inhomogeneous system, solve the homogeneous system and add a particular solution to it. You can solve it by guessing and testing or by the fundamental matrix Φ composed of the solutions to the homogeneous system

$$x_1 = e^{it}\begin{bmatrix} \frac{-1-i}{2} \\ 1 \end{bmatrix}, x_2 = e^{-it}\begin{bmatrix} \frac{-1+i}{2} \\ 1 \end{bmatrix}$$

$$\Phi = \begin{bmatrix} e^{it}(\frac{-1-i}{2}) & e^{-it}(\frac{-1+i}{2}) \\ e^{it} & e^{-it} \end{bmatrix}$$

The general real solution to a linear inhomogeneous system is:

$$x(t) = \Phi(t)c_{1,2\ldots n} + \Phi(t)\int_{t_o}^{t} [\Phi(s)]^{-1} u(s)\, ds$$

The particular solution for which x(t_0) = x_0 is given by

$$x(t) = \Phi(t)[\Phi(t_0)]^{-1}x_0 + \Phi(t)\int_{t_o}^{t} [\Phi(s)]^{-1} u(s)\, ds$$

For a 2x2 matrix A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \cdot \frac{1}{det(A)}$

In the case of a special system $\begin{cases} \underline{x}' = A\underline{x} + \underline{b}u \\ y = \underline{d}^T \underline{x} = d_1 x_1 + d_2 x_2 + \ldots + d_n x_n \end{cases}$, $H(s) = -d^T(A - sI)^{-1}b$

Exactly as before, if $u(t) = e^{st}, y(t) = H(s)e^{st}$. If $u(t) = cos(wt)$ or $sin(wt)$, apply the Re() and Im() rules on the previous page.

$\underline{x}'(t) = Ax(t)$ is stable if and only if:

1. No λ has a positive real part
2. Every λ with a real part of 0 has equal algebraic and geometric multiplicity

It is asymptotically stable if and only if:

1. All λ have a negative real part

If the polynomial is in the form $\lambda^2 + a_1\lambda + a_2$, the λ have negative real parts ifa_1, a_2 are positive

If the polynomial is in the form $\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$, the λ have negative real parts if a_1, a_2, a_3 are positive and $det(\begin{bmatrix} a_1 & a_3 \\ 1 & a_2 \end{bmatrix})$ is positive

An inhomogeneous system is asymptotically stable and BIBO stable if and only if the associated homogeneous system is asymptotically stable

2 Sequences and Series

2.1 Convergence Tests

The series $\sum_0^\infty x^n$ is called a geometric series and has a sum of $\frac{1-x^n}{1-x}$ where n is the term you begin with. If n is 0, the sum is $\frac{1}{1-x}$. If $|x| < 1$, the series converges. Otherwise it diverges

You can freely change the index by the following principle: $\sum_0^\infty a_n = \sum_N^\infty a_{n-N}$

A series is called conditionally convergent if it converges but its absolute value counterpart does not. If it does, it is called absolutely convergent.

Quotient Test: if $|\frac{a_{n+1}}{a_n}| < 1$, as n tends to ∞, the series converges absolutely

Comparison Test: if $0 \leq a_n \leq b_n$, if $\sum a_n$ diverges, $\sum b_n$ diverges. If $\sum b_n$ converges, $\sum a_n$ converges

Equivalence Test: If $\frac{a_n}{b_n}$ tends towards a constant as n tends to infinity, both series will behave the same. Either both converge or both diverge.

Liebnitz's Test: For an alternating series $\sum_1^\infty (-1)^n b_n$, if b_n > 0 for all n ∈ N, is monotone decreasing, and tends to 0 as n tends to infinity, then the series is at least conditionally convergent. If $\sum_1^\infty |a_n|$ is convergent, $\sum_1^\infty a_n$ is absolutely convergent.

Integral Test: If $\sum_1^\infty a_n$ is positive, continuous, and decreasing, if $\int_1^\infty a_n$ converges (evaluates to a number), then $\sum_1^\infty a_n$ converges.

2.2 Sum Approximation

To approximate $\sum_1^\infty f(n)$ with an error at most ϵ:

1. find $\int_N^\infty f(x)dx < \epsilon$,or
2. find $f(N+1) < \epsilon,\ S \approx S_N + \int_{N+1}^\infty f(x)$

Ex: Approximate to an error < 0.01

$$\sum_1^\infty \frac{1}{n^3}$$

$$1.\ \int_N^\infty \frac{1}{x^3}\ dx = \frac{1}{2N^2},\ \frac{1}{2N^2} < 0.01,\ N > \sqrt{50} \approx 7.07$$

8 is the next integer

$$\sum_1^\infty \frac{1}{n^3} \approx S_8$$

$$2.\ \frac{1}{(N+1)^3} < 0.01,\ N > 100^{1/3} - 1 \approx 3.64$$

$$\sum_1^\infty \frac{1}{n^3} \approx S_4 + \int_5^\infty f(x)\ dx$$

2.3 Infinite Series

A Power Series is in the form $\sum_0^\infty c_n(x-x_0)^n$ or $\sum_0^\infty c_n x^n$ A Power Series can either converge only for x = 0, converge absolutely (regardless of x), or there is a constant p such that the series is only convergent for $|x| < p$. The value p is called the radius of convergence. It is almost always found through the quotient test, where $|\frac{a_{n+1}}{a_n}|$ will end up depending on x. Upon finding p, you must check the convergence at p and -p manually, called the boundary, as the quotient test does not prove convergence exactly at ± p

A Taylor Series approximation can be created for an arbitrarily often differentiable function.

$$f(x) = \sum_{0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

A series $\sum_1^\infty f_n(x)$ can be discontinuous even if all $f_n(x)$ are continuous

Pointwise convergence means for any error ϵ, you can find $|f(x) - S_N(x)| < \epsilon$, for every x individually

Uniform convergence means for any error ϵ, you can find $|f(x) - S_N(x)| < \epsilon$, which will work for all x regardless of its value

A series $\sum_1^\infty k_n(x)$ is called a majorant series for $\sum_1^\infty f_n(x)$ if $|f_n(x)| \leq k_n$ for all relevant x and n. The majorant series cannot depend on x. If the majorant series converges, it is a convergent majorant series. If a series has a convergent majorant series, then it itself is uniformly convergent. If all f_n are continuous and the series $\sum_1^\infty f_n(x)$ is uniformly convergent, then it is also continuous.

2.4 Fourier Series

A Fourier Series can be defined for a periodic, bounded function f where:

$$f \approx \frac{1}{2}a_0 + \sum_{1}^{\infty}(a_n cos(nx) + b_n sin(nx))$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\ dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)cos(nx)\ dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)sin(nx)\ dx$$

A function is piecewise differentiable if it can be divided into a finite number of sections so that the pieces of the function are differentiable on those sections, and the derivative of the function on those sections is continuous

If a function is 2π periodic and piecewise differentiable, then its fourier series is convergent

At a point where the fourier series is not continuous, its value at the point is $\frac{f(x^+)+f(x^-)}{2}$

If a function is 2π periodic, piecewise differentiable, and continuous, then the fourier series of the function is equal to the function. and $|f(x) - S_N(x)| \leq \sqrt{\frac{1}{N\pi}\int_{-\pi}^{\pi}|f'(x)|^2\ dx}$, which is the error.

In a more practical form, to find N for given ϵ, $N \leq \frac{1}{\epsilon^2\pi}\int_{-\pi}^{\pi}|f'(x)|^2\ dx$

Alternatively, $|f(x) - S_N(x)| \leq \sum_{N+1}^{\infty}(|a_n| + |b_n|)$

If the function is even (symmetric around the y axis):

$$a_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)cos(nx)\ dx$$

$$b_n = 0$$

If the function is odd (symmetric around the origin):

$$a_n = 0$$

$$b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)sin(nx)\ dx$$

Disclaimer: From my understanding the general practice during the course was to take out the c_0 term out of the sum, as it does not depend on anything. The book usually keeps c_0 as just part of the summation $\sum_{-\infty}^{\infty}$. As far as I'm aware this shouldn't change any of the calculations. For completeness' sake, this document usually takes out the c_0 and states $n \neq 0$

The complex fourier coefficient cn is:

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx}$$

In complex form, you can write the fourier series as:

$$f \approx c_o + \sum_{-\infty}^{\infty} c_n e^{inx}, \ n \neq 0$$

$$c_0 = \frac{a_0}{2}$$

$$c_n = \frac{a_n - ib_n}{2}, \ n \in 1, 2, 3, \ldots$$

$$c_n = \frac{a_{-n} + ib_{-n}}{2}, \ n \in -1, -2, -3, \ldots$$

To go from complex to real fourier coefficients:

$$a_n = c_n + c_{-n}$$

$$b_n = i(c_n - c_{-n})$$

If for any reason you solve the error in the complex form, $|f(x) - S_N(x)| \leq \sum_{N+1}^{\infty}(|c_n|)$. Usually this will require hyperbolic trigonometry.

$$cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$cos(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

2.5 Fourier's Method

Consider a first order system of Diff. Eq. $\begin{cases} \underline{x}' = A\underline{x} + \underline{b}u \\ y = \underline{d}^T \underline{x} = d_1 x_1 + d_2 x_2 + \ldots + d_n x_n \end{cases}$, $H(s) = -d^T(A - sI)^{-1}b$, where u is a 2π periodic function. The procedure for Fourier Method is:

1. Check if the system is asymptotically stable
2. Find the transfer function $H(s) = -d^T(A - sI)^{-1}b$
3. Check that u is piecewise differentiable, then find its fourier series
4. Check that u is continuous, $u(t) = c_o + \sum_{-\infty}^{\infty} c_n e^{int}, \ n \neq 0$
5. $y(t) = c_0 H(0) + \sum_{-\infty}^{\infty} c_n H(in) e^{int}, n \neq 0$
6. If asked, find $|y(t) - \sum_{-N}^{N} c_n H(in) e^{int}$ by using $y(t)$ in series form. Effectively, $|y(t) - \sum_{-N}^{N} c_n H(in) e^{int} \leq \sum_{|n| \geq N+1} |c_n||H(in)|$, where after evaluating the right hand side you can use the first solution approximation method stated in a section above

2.6 Power Series Method

$$y(t) = \sum_0^\infty c_n t^n$$

$$y'(t) = \sum_1^\infty n c_n t^{n-1}$$

$$y''(t) = \sum_2^\infty n(n-1) c_n t^{n-2}$$

$$y^{(k)}(t) = \sum_k^\infty n(n-1)\dots(n-k+1) c_n t^{n-k}$$

$$y^{(k)}(0) = k! c_k,\ 0! = 1$$

To change index, and this also works when you want to increase the index:

$$\sum_N^\infty c_n t^n = \sum_0^\infty c_{n+N} t^{n+N}$$

Power Series Method for solving Diff. Eq.:

1. Assume a solution in the form $y(t) = \sum_0^\infty c_n t^n$
2. Insert this $y(t)$ into the Diff. Eq. and find the coefficients c_n through aligning the power of t and the index of the series'
3. (Quite often asked for) Find radius of convergence through Quotient test

Ex:

$$ty'' - ty' - y' = 0$$

$$y(t) = \sum_0^\infty c_n t^n$$

$$t\left(\sum_2^\infty n(n-1) c_n t^{n-2}\right) - t\left(\sum_1^\infty n c_n t^{n-1}\right) - \sum_0^\infty c_n t^n = 0$$

$$\sum_2^\infty n(n-1) c_n t^{n-1} - \sum_1^\infty n c_n t^n - \sum_0^\infty c_n t^n = 0$$

First align the powers by changing the index

$$\sum_1^\infty (n+1)(n) c_{n+1} t^n - \sum_1^\infty n c_n t^n - \sum_0^\infty c_n t^n = 0$$

Separate out a term to fix the index

$$\sum_1^\infty (n+1)(n) c_{n+1} t^n - \sum_1^\infty n c_n t^n - c_0 - \sum_1^\infty c_n t^n = 0$$

$$-c_0 + \sum_{1}^{\infty}((n+1)(n)c_{n+1} - nc_n - c_n)t^n = 0$$

$$-c_0 = 0, (n+1)(n)c_{n+1} - (n+1)c_n = 0$$

$$c_{n+1} = \frac{n+1}{n(n+1)}c_n = \frac{1}{n}c_n$$

$$c_n = \frac{1}{(n-1)!}c_1,\ n \geq 2$$

(I'm not explaining the math for that step as it's just part of the solving of this specific example and not something relevant as a general rule)
No condition on $c_1$so it's a free parameter

$$y(t) = c_1(t) + \sum_{2}^{\infty}\frac{1}{(n+1)!}c_1t = \sum_{1}^{\infty}\frac{1}{(n+1)!}c_1t$$

$$|\frac{c_{n+1}x^{n+1}}{c_nx^n}| = |\frac{\frac{1}{n}c_nt^{n+1}}{c_nt^n}| \to |\frac{1}{n}t| \to 0$$

So radius of convergence $p = \infty$ as $0 < 1$ regardless of t

For inhomogeneous equation it's basically the same process:

$$ty'' + 2ty' + y = \frac{1}{1+t} = \sum_{0}^{\infty}(-1)^nt^n$$

$$\dots$$

$$c_0 + 3c_1 + \sum_{2}^{\infty}(n^2+n+1)c_nt^n = \sum_{0}^{\infty}(-1)^nt^n = 1 - t + \sum_{2}^{\infty}(-1)^nt^n$$

$$c_0 = 1, c_1 = \frac{-1}{3}, c_n = \frac{(-1)^n}{n^2+n+1}, n \geq 2$$